

Harsha Venkateshwara

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Summary

AI engineer who turned a **4-day enterprise workflow into a 90-second agentic pipeline** at Commvault - then shipped production ML at **8–10M+ record scale** at Itron, won a **10K hackathon**, and earned back-to-back **4.92/5 client ratings**. Building the same production rigor into AI systems at UB.

Technical Skills

Programming & Data: Python, C++, NumPy, Pandas, Scikit-Learn, OpenCV, Linux, Git

ML Frameworks: PyTorch, JAX/Flax, TensorFlow, Hugging Face, vLLM, TensorRT, Ray, DeepSpeed, FSDP, Orbx, Grain

GPU & Systems: CUDA C/C++, Triton, NCCL, MPI, XLA, Nsight Systems/Compute, Kernel Fusion, Memory Optimization

Cloud & Infra: AWS, GCP, Azure, Docker, Kubernetes, Slurm, ROS2, Autoware, Digital Twin, CI/CD, TensorBoard

Engineering: Distributed Systems, MLOps, Data Engineering (ETL), OOP, Data Structures & Algorithms, Agile

Experience

Commvault

AI Engineer Intern

May 2026 – Present

- Engineered SOF-Forge, an agentic **RAG** pipeline using **GPT-4.1**, **LangGraph**, and **Azure OpenAI**, automating Summary of Findings generation from **Dynamics 365** CRM data - cutting documentation time from **4 days to under 90 seconds**.
- Architected a multi-agent system with PII/PHI sanitization guardrails, **Cosmos DB** persistence, and **Azure Blob Storage** delivery via **FastAPI** - achieving **zero data leakage** across **100%** of validated outputs.
- Designed a dual-path LLM classification engine with prompt engineering and RAG across six parallel section generators, delivering audit-ready Word documents via SAS-authenticated APIs with latency **under 30 seconds**.

Research Foundation for SUNY - CAVAS Lab

Graduate Research Assistant, Autonomous Systems

Jan 2026 – May 2026

- Investigated perception and planning architectures for autonomous vehicles using **ROS2** and **Autoware** within physics-accurate **digital twin** simulation environments, contributing to real-world deployment research.

Itron, Inc.

Technical Associate, Machine Learning & Data Systems

Aug 2023 – Aug 2025

- Architected production ML pipelines for MV-90xi MDM ingesting **8–10M+** smart-meter records for load forecasting and anomaly detection, achieving **18–30%** accuracy gains - recognized with **2 SPOT Awards** and a sustained **4.92/5.0 CSAT rating** over 2 consecutive years.
- Slashed data noise by **25%** through feature engineering and pipeline optimization, sustaining **90%+** model accuracy under continuously shifting production distributions.
- Automated ingestion, training, and retraining workflows in Python - cutting manual effort by **35%** and doubling experimentation velocity across million-record operational environments.

Kyono Software Pvt. Ltd.

Machine Learning Engineer

Jun 2022 – Jul 2023

- Spearheaded end-to-end ML delivery across **6–8 client engagements** in e-commerce, healthcare, and retail - owning full model lifecycle from data preparation through deployment and post-launch monitoring.
- Designed recommendation and classification systems lifting outcomes by **10–20%**; built preprocessing pipelines handling **100K–1M+** records from heterogeneous data sources.

Bharat Electronics Limited

Data Science Intern - Analytics & Reporting, Air Force Defense Unit

Sep 2021 – Oct 2021

- Analyzed and visualized sensitive operational datasets for Indian Air Force defense systems in a mission-critical environment, maintaining **100%** security, audit, and documentation compliance.

Projects

TVLift — Convergent TV Attribution Engine | GitHub

- Built a production-grade Media Mix Model using geo-lift causal inference, Bayesian MMM (PyMC), and Hill-adstock saturation; deployed XGBoost + SHAP attribution and a budget optimizer across multi-channel TV, social, and search spend.

AEGIS — AI Support Automation Engine | GitHub | Live Demo

- Built a production-grade agentic support automation platform using a 5-layer intent router (regex → FAISS RAG → Groq Llama 3.3 70B), streaming chatbot, Gmail IMAP/SMTP automation, SQLite CRM, and Plotly analytics dashboard — covering 12 intents, achieving **90%+** rule-based resolution with automatic ticket escalation and SLA tracking across P1–P4 tiers.

Electric Smart Grid Stabilizer | Itron, Inc. Hackathon | GitHub

- Engineered an end-to-end ML pipeline for telemetry normalization, feature engineering, and label construction; benchmarked Scikit-Learn against PyTorch deep learning models for real-world grid deployment. **Awarded \$10K cash prize.**

LithoStream Sentinel — Real-Time Machine Monitoring Platform | GitHub

- Implemented a real-time telemetry pipeline ingesting machine signals via FastAPI into InfluxDB, monitored with Prometheus, and visualized in live Grafana dashboards; containerized end-to-end with Docker Compose.

Education

University at Buffalo, The State University of New York

Buffalo, NY

Master of Science, Computer Science and Engineering

Aug 2025 – Dec 2026

- Coursework:** Algorithm Analysis & Design, Machine Learning, Deep Learning, Statistical Data Mining II, Data Intensive Computing, Data Models & Query Language, Operating Systems, Computer Security

Visvesvaraya Technological University

India

Bachelor of Engineering, Mechanical Engineering

Jun 2018 – Jul 2022